



STUDENT COURSE COMPLETION PREDICTION & DECISION SUPPORT SYSTEM.

Translating Data Science into Actionable EduTech Solutions

IT3081 Statistical Modelling

Presented by:

2026-DS-49

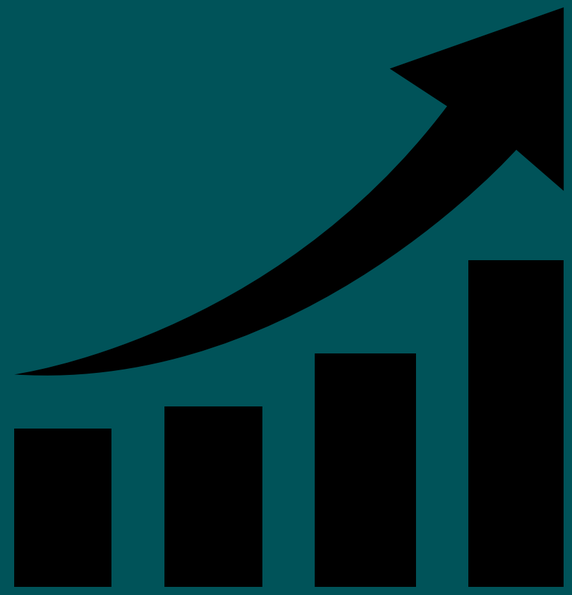
Team Members

IT Number	Name
IT24101896	Abeywickrama J B
IT24101520	Athukorala A P H B
IT24101469	Bandara W M S S
IT24101376	Nishshanka N P S M



Presentation Roadmap

- Industry problem and consultancy objectives
- Dataset, analysis, modelling, and evidence
- Innovation proposal and final recommendations
- Industry validation



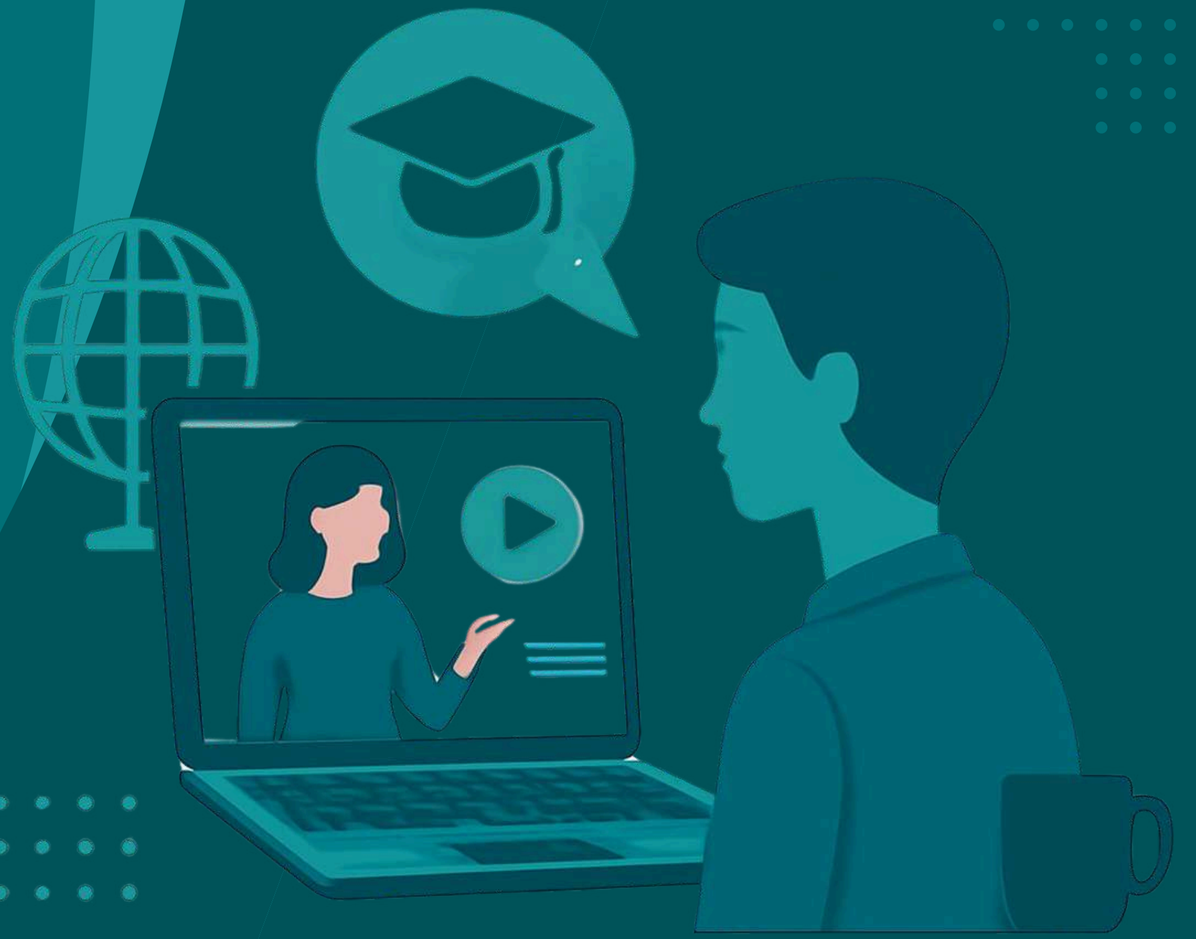
Industry problem and consultancy objectives





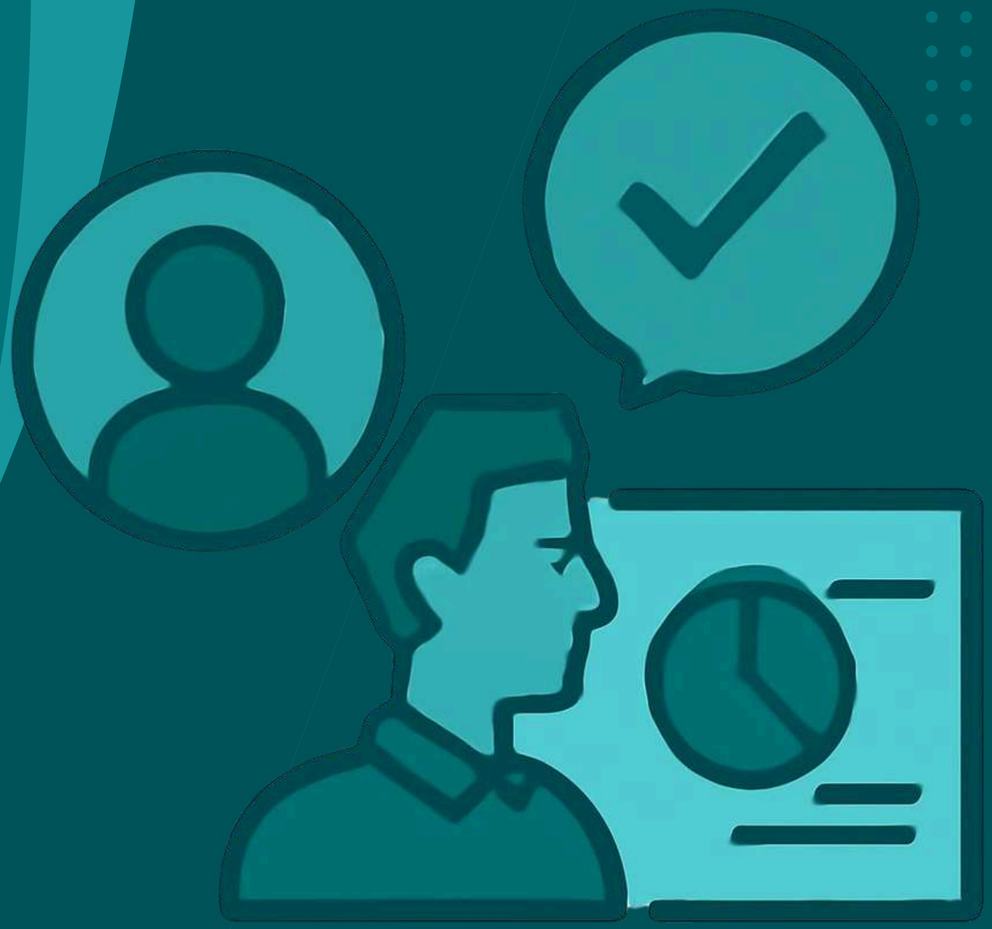
Industry background and organizational context

- **Digital Education**: Universities increasingly use LMS platforms for online learning.
- **Student Data**: LMS systems generate large amounts of student activity data.
- **Need for Action**: Universities use data to improve student retention and success.



Stakeholders affected by the problem

- University senior management
- Academic advisors
- Enrolled students



Industry problem and consultancy objectives



Consultancy Objectives

- Predict student course completion accurately.
- Create an Early Warning System and Advisor Dashboard to identify and support at risk students early.

Industry Problem

Higher education platforms and online learning ecosystems suffer from high student dropout rates. Teachers and management find out about student failures too late to provide support.

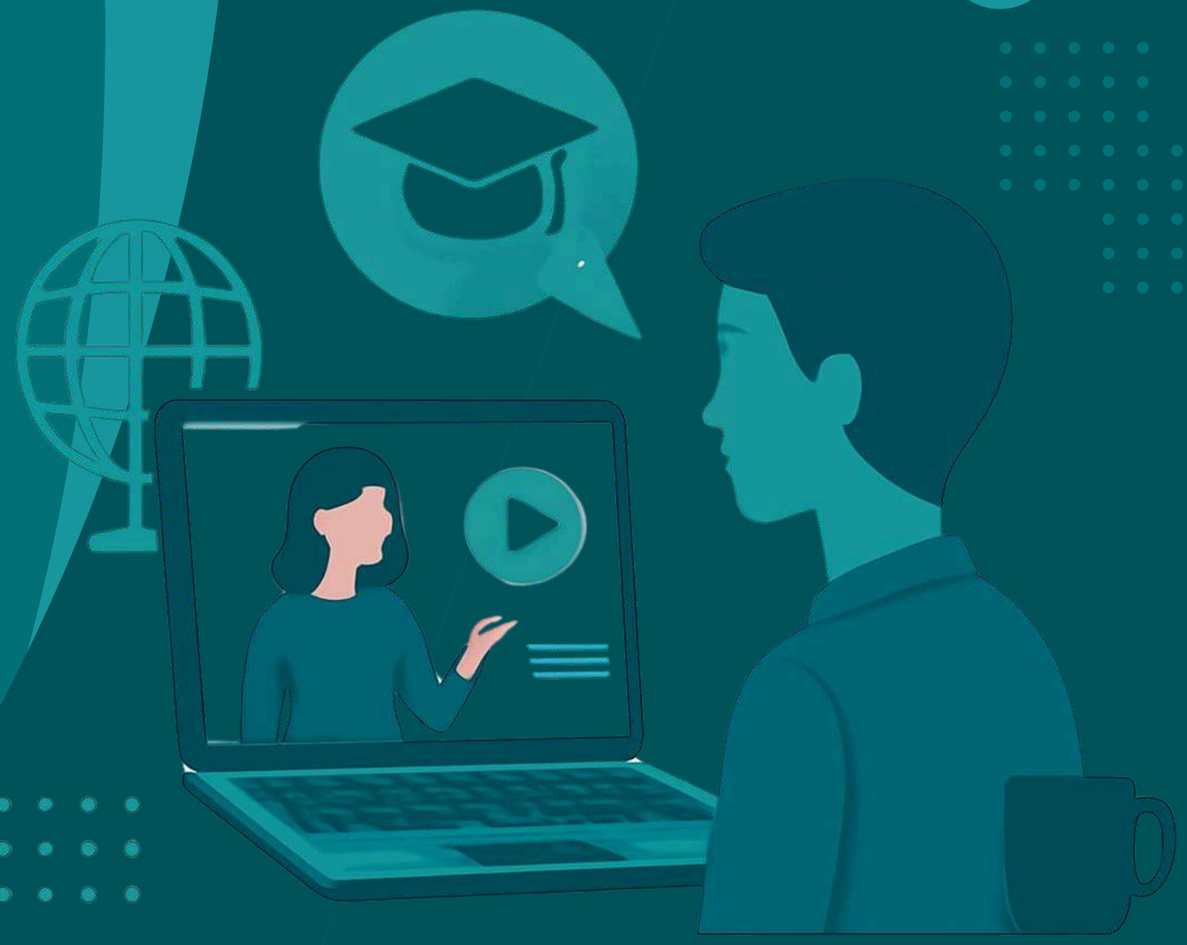
Research Landscape & Research Gap





Current industry practices and statistical approaches

- **Traditional Approach**: Post semester grades and manual student tracking.
- **Statistical Approach**: Mainly basic reports and descriptive analysis.
- **Limitation**: Limited use of predictive models for early intervention.



Research Support: Findings are supported by educational data mining research (Baker & Inventado, 2014).

Strengths and weaknesses of previous studies

Strengths: Identified key factors affecting student performance.

Weaknesses: Mostly descriptive, post semester analysis with limited real-time support.



Research Gap & Justification

- **Research Gap:** Lack of transparent predictive systems with actionable advisor support.
- **Our Approach:** Our analysis uses interpretable Ridge Regression and Bayesian analysis for early student risk detection.



Dataset Selection & Understanding





Dataset source and public accessibility

- **Source**: Kaggle – Student Course Completion Prediction dataset.
- **Access**: Publicly available open-access dataset.

Variables and target outcome

- **Dataset**: 100,000 student records with 40 variables.
- **Key Features**: Progress, study time, quiz scores, and login activity.
- **Target**: Records split into “completed”(51.87%) vs “dropouts (48.13%).

Data quality, Missing values, and Outliers

- **Data Quality**: 100,000 complete records with 40 attributes.
- **Missing Values**: 0 missing values.
- **Outliers**: Scalled down using log transformation and removed some.
- **Feature Engineering**: New features created Total_Engagement_Score, Inactivity_Penalty_Ratio, log_Time_Spent and Paid_Engagement.

Why the dataset supports business decisions?

- **Realistic Data**: Captures students' daily learning behaviors.
- **Early Detection**: Helps identify students at risk of dropping out through pattern..
- **Business Value**: Supports stakeholders to retain students throughout courses, better feedback on study material and promoting courses to students.

Dataset

```
> nrow(student_data)
[1] 100000
> ncol(student_data)
[1] 40
> head(student_data)
  Student_ID            Name Gender Age Education_Level Employment_Status    City Device_Type
1  STU100000    Vihaan Patel   Male  19         Diploma           Student  Indore      Laptop
2  STU100001      Arjun Nair Female  17        Bachelor           Student   Delhi      Laptop
```

Checking Missing Data

```
> missing_counts <- colSums(is.na(student_data))
> print(missing_counts)
Student_ID                       Name                     Gender 
         0                          0                          0 
       Age            Education_Level          Employment_Status 
         0                          0                          0 
```

```
> missing_percentage <- (colSums(is.na(student_data)) / nrow(student_data)) * 100
> print(missing_percentage)
Student_ID                       Name                     Gender 
         0                          0                          0 
       Age            Education_Level          Employment_Status 
         0                          0                          0 
```

Descriptive Analysis & Initial Insights





Important descriptive statistics

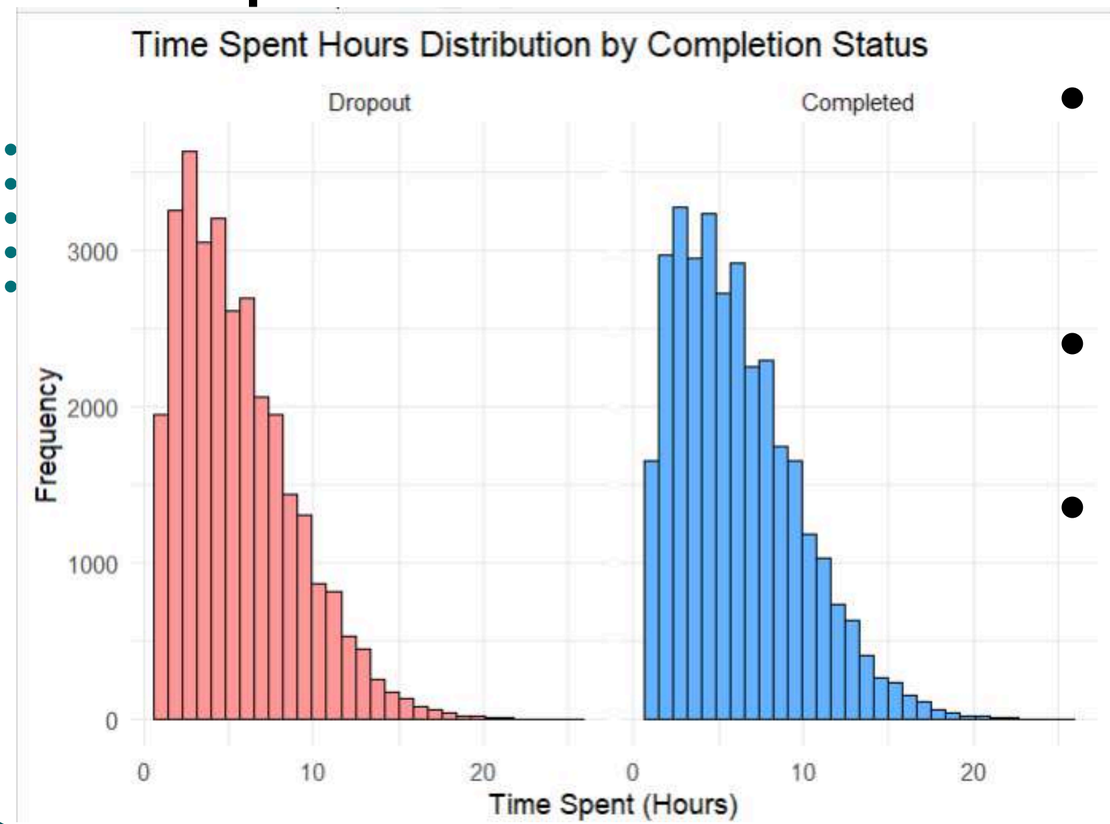
- Study Time: Average = 5.67 hours (SD = 3.2528)
- Quiz Score: Average = 73.79 (SD = 12.3537)
- Overall: Students show moderate study time and quiz performance.

R output

```
  Mean_Time_Spent SD_Time_Spent Mean_Quiz_Score SD_Quiz_Score
1        5.668588      3.252816        73.78737      12.35366
>
```

Key graphs and patterns

R output



- **Quiz scores** show a right-skewed distribution, with most students scoring between 60 and 80.
- **Study time** is a key indicator of course completion.
- **Completed students** spend **more active** study time than students who drop out.

Initial business insights

- **Warning Signs**: Long login gaps and low video completion indicate disengagement.

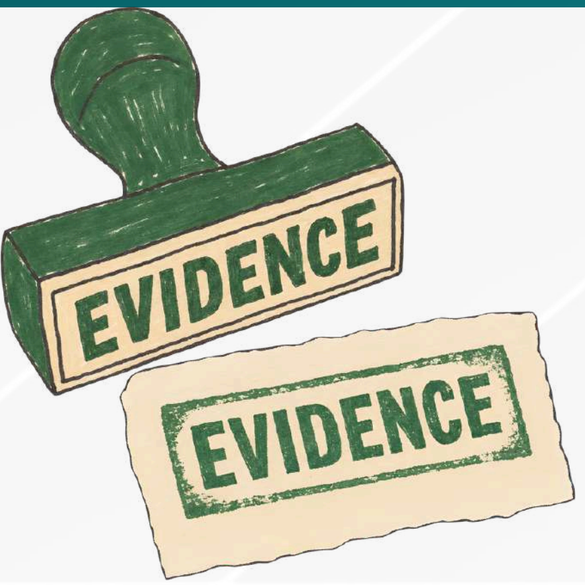
- **Study Habits**: Successful students show consistent platform activity.

What management can learn from the data?

- Proactive Retention: Ongoing engagement is a key indicator of completion.

- Early Intervention: Login frequency and study time can help identify at risk students early.

Statistical Inference Evidence





Research Questions & Hypotheses

Question

Do student study hours, demographics, and technical access factors significantly influence course completion?



Hypothesis

Study duration,
employment status,
and device access

have a significant relationship with student retention.

Selected Tests & Justification

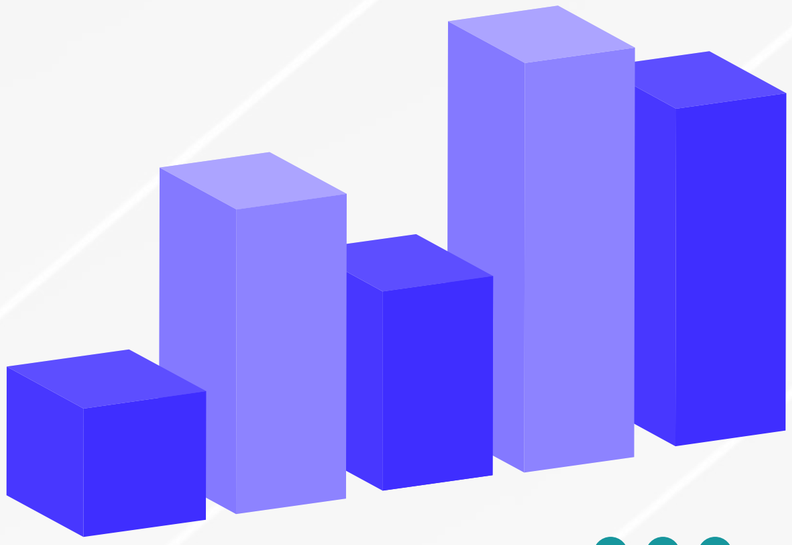
Tests

- **t-Test**
- **F-Test**
- **Chi-Square**
- **ANOVA**

Justifications

- **t-Test & F-Test**: Compare study time means and variances between completers and dropouts.
- **Chi-Square**: Test relationships between completion and Employment, Gender, and Device Type.
- **ANOVA**: Compare average quiz scores across different course levels.

Main Statistical Results



Implementation

- Time Spent (**t-test**):
  t = t=17.014, $p<2.2 \times 10^{-16}$
- Employment Status (**Chi-Square**):
  $x^2 = 31.836$, $p = 5.668 \times 10^{-7}$
- Device Type (**Chi-Square**):
  $x^2 = 1.7639$, $p = 0.0414$
- Quiz Score vs Course Level (**ANOVA**):
  $F = 4.207$, $p = 0.0149$

Hypothesis Test - Two Sample t-test (Comparison of Means)

```
        Welch Two Sample t-test

data:  Time_Spent_Hours by Completed
t = 17.014, df = 57595, p-value < 2.2e-16
alternative hypothesis: true difference in means between group Completed and group Not Completed is not equal to 0
95 percent confidence interval:
 0.4064764 0.5123231
sample estimates:
    mean in group Completed mean in group Not Completed
                   5.889678                    5.430278
```

Hypothesis Test - Comparison of Variances (F-Test)

```
        F test to compare two variances

data:  Time_Spent_Hours by Completed
F = 1.0917, num df = 29906, denom df = 27745, p-value = 1.019e-13
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 1.066775 1.117227
sample estimates:
ratio of variances
          1.091715
```

ANOVA Test (Comparison of Means across Multiple Groups)

```
> summary(aov(Quiz_Score_Avg ~ Course_Level, data = student_data))
                Df  Sum Sq Mean Sq F value Pr(>F)
Course_Level     2    1284   642.0   4.207 0.0149 *
Residuals    57650 8797149   152.6
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Comparison of Proportions (Chi-Square Test for Gender)

```
        Pearson's Chi-squared test

data:  student_data$Completed and student_data$Gender
X-squared = 6.8567, df = 2, p-value = 0.03244
```

Chi-Square Test for Course Completion vs Device Type

```
        Pearson's Chi-squared test

data:  student_data$Completed and student_data$Device_Type
X-squared = 1.7639, df = 2, p-value = 0.414
```

Chi-Square Test for Course Completion vs Employment Status

```
        Pearson's Chi-squared test

data:  student_data$Completed and student_data$Employment_Status
X-squared = 31.836, df = 3, p-value = 5.668e-07
```

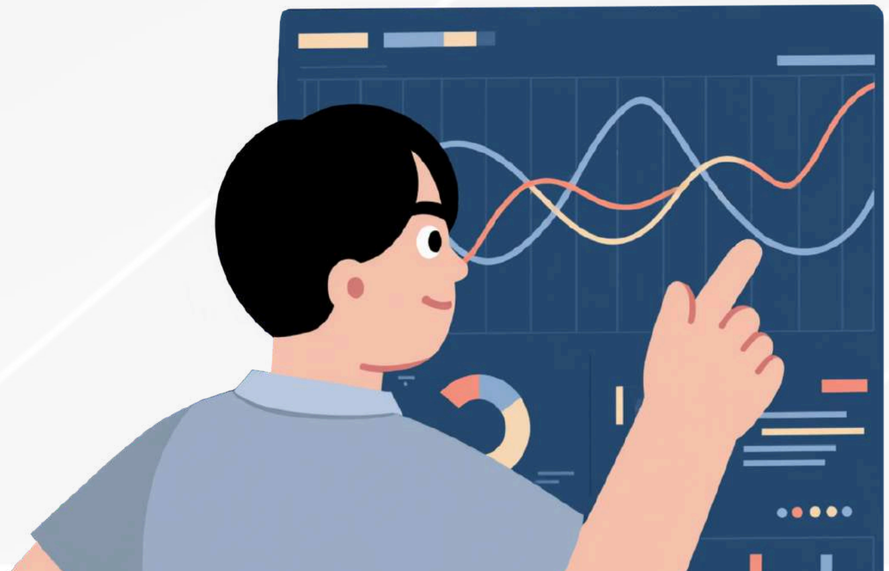
Practical implication for the organization

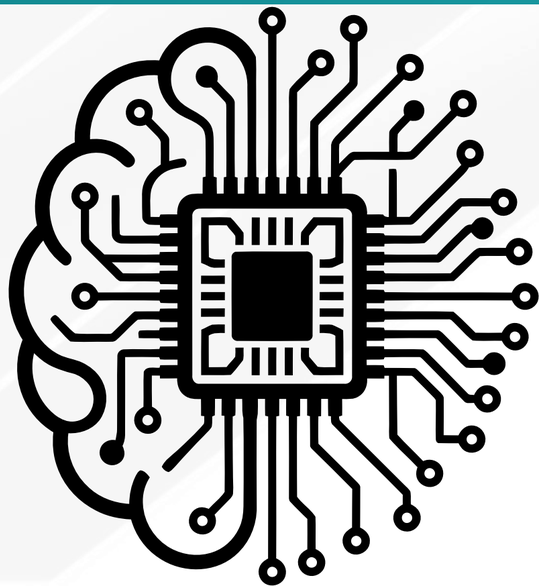


Practical Implication

- **Personalized Retention**: Consider students' employment and device access.
- **Early Warning**: Use active study time to identify at-risk students.

Predictive Statistical Modelling





Model Selection & Evaluation

Selected Models

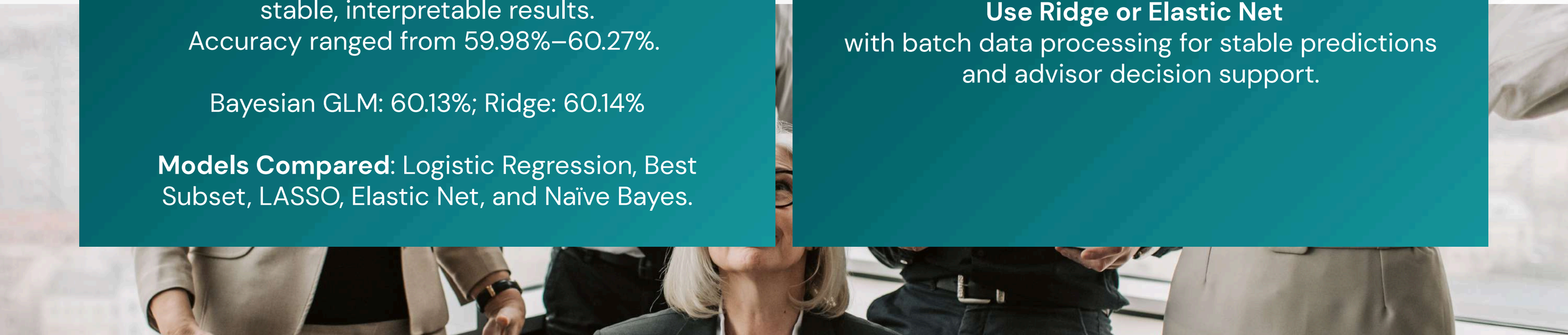
Ridge Regression and Bayesian Logistic Regression to handle multicollinearity and provide stable, interpretable results.
Accuracy ranged from 59.98%–60.27%.

Bayesian GLM: 60.13%; Ridge: 60.14%

Models Compared: Logistic Regression, Best Subset, LASSO, Elastic Net, and Naïve Bayes.

Recommended model

Use Ridge or Elastic Net
with batch data processing for stable predictions and advisor decision support.



Modelling - LASSO Regression

```
> # LASSO (Alpha = 1)
> set.seed(123)
> cv_lasso    <- cv.glmnet(x_train, y_train, family = "binomial", alpha = 1)
> lasso_preds <- predict(cv_lasso, newx = x_test, s = "lambda.min", type = "response")
> lasso_class <- ifelse(lasso_preds > 0.5, 1, 0)
> acc_lasso   <- (sum(diag(table(Predicted = lasso_class, Actual = y_test))) / length(y_test)) * 100
>
> cat("LASSO Regression:", round(acc_lasso, 2), "%\n")
LASSO Regression: 60.25 %
```

Modelling - Elastic Net

```
> # Elastic Net (Alpha = 0.5)
> set.seed(123)
> cv_enet    <- cv.glmnet(x_train, y_train, family = "binomial", alpha = 0.5)
> enet_preds <- predict(cv_enet, newx = x_test, s = "lambda.min", type = "response")
> enet_class <- ifelse(enet_preds > 0.5, 1, 0)
> acc_enet   <- (sum(diag(table(Predicted = enet_class, Actual = y_test))) / length(y_test)) * 100
>
> cat("Elastic Net Regression:", round(acc_enet, 2), "%\n")
Elastic Net Regression: 60.27 %
```

Summary of Evidence

EDA Summary & Outputs

- Total Rows: 100,000
- Total Variables: 43
- Missing Values: 0%
- Target Variable Split:

Dropouts (0) vs Completed (1).

Hypothesis Testing Summary

t-test: $t = 17.014$, $p < 2.2 \times 10^{-16}$

ANOVA test: (avg quiz levels are different or not)
$F = 4.207$, $p = 0.0149$
(significant)

F-Test: $F = 1.0917$, $p < 1.019 \times 10^{-13}$

Chi-Square (Completion vs Gender):
$x^2 = 6.856$, $p = 0.032$

Chi-Square (Completion vs Employment):
$x^2 = 31.836$, $p < 5.668 \times 10^{-7}$

Chi-Square (Completion vs Device Type):
$x^2 = 1.763$, $p = 0.414$

Model Comparison & Accuracies

Bayesian Logistic Regression – 60.13%

LASSO Regression ($\alpha = 1$) – 60.25%

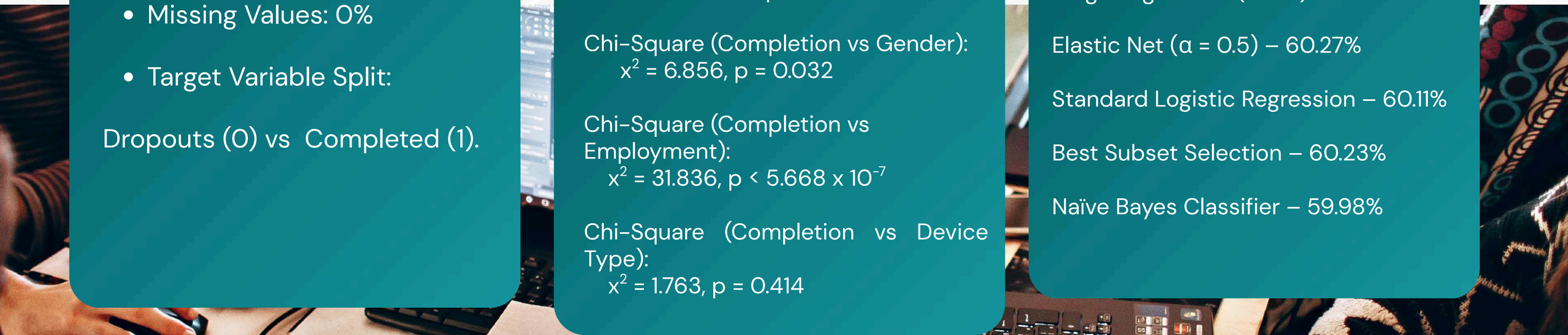
Ridge Regression ($\alpha = 0$) – 60.14%

Elastic Net ($\alpha = 0.5$) – 60.27%

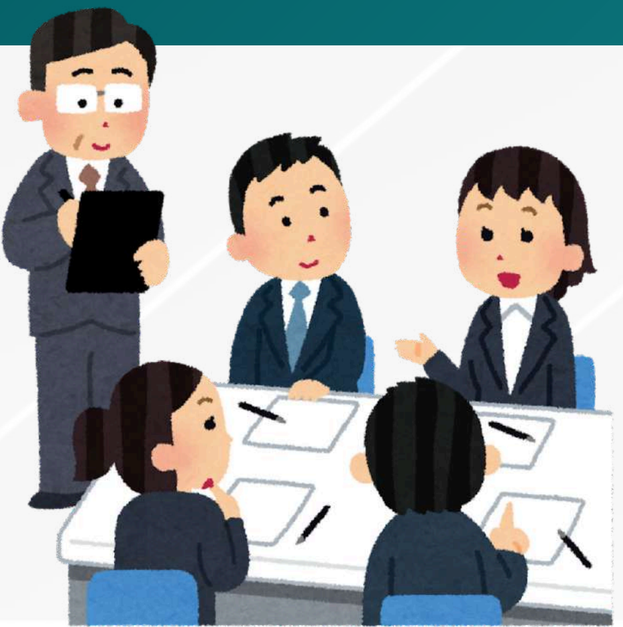
Standard Logistic Regression – 60.11%

Best Subset Selection – 60.23%

Naïve Bayes Classifier – 59.98%



Experimental Design Evaluation





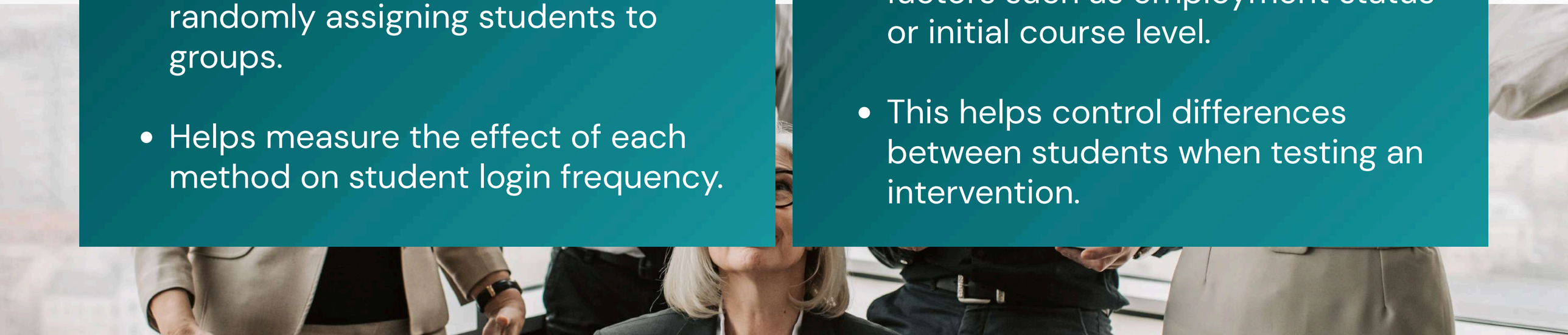
Possible use of CRD & RCBD

CRD (Completely Randomized Design)

- Can test different reminder methods, such as SMS vs. email, by randomly assigning students to groups.
- Helps measure the effect of each method on student login frequency.

RCBD (Randomized Complete Block Design)

- Can group students based on factors such as employment status or initial course level.
- This helps control differences between students when testing an intervention.



Advantages, Limitations & Feasibility

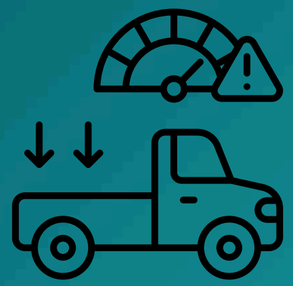
Advantages

Provides stronger evidence about whether an intervention actually causes better student retention.



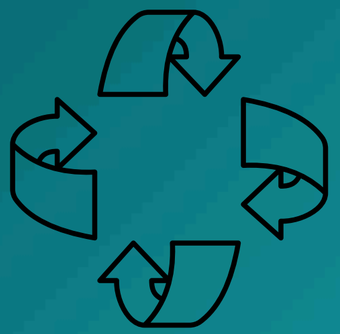
Limitations

Experiments can be difficult to manage during a live semester and may create fairness concerns.



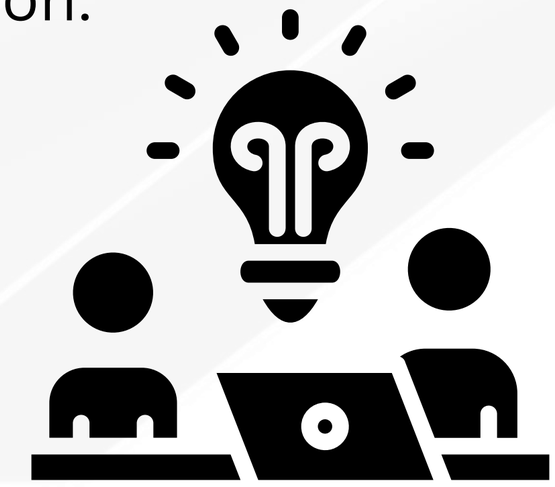
Feasibility

Can be implemented as a controlled pilot with selected student groups.



Future Decision Support

- Experimental results can help improve EduPulse AI intervention triggers.
- This allows the institution to focus advisor time and automated alerts on interventions that show measurable improvements in retention.



PCA Evaluation





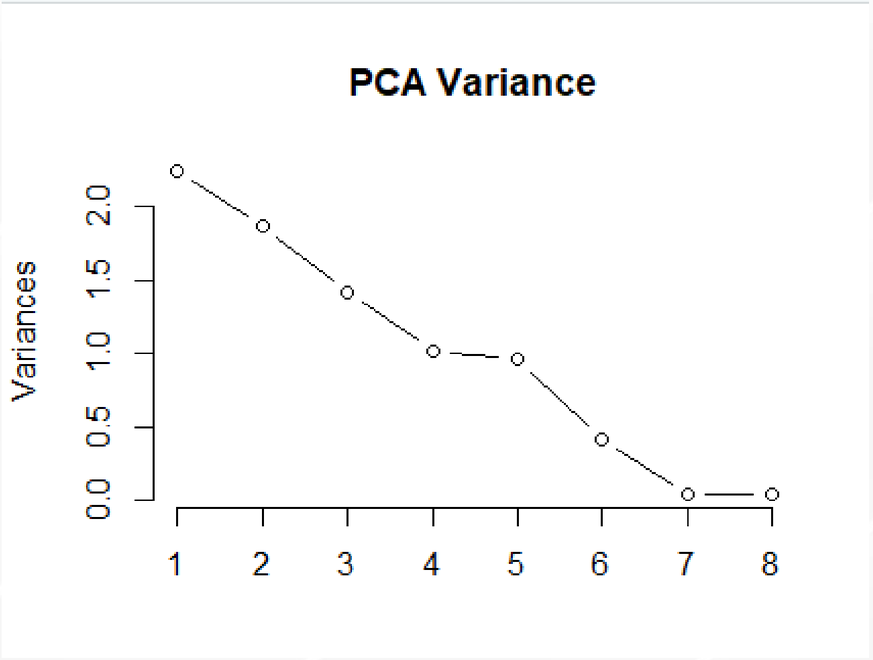
PCA Findings & Model Impact

Need

Dimensionality reduction is unnecessary because regularized models handle multicollinearity.

Limitation

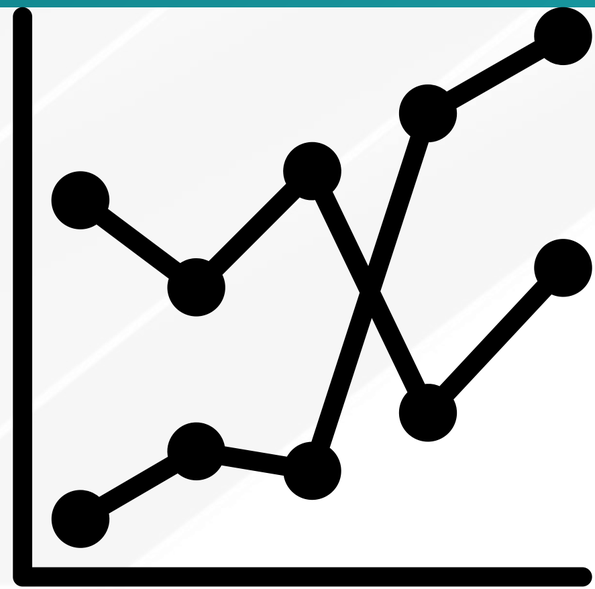
PCA reduces interpretability by replacing meaningful features with abstract components.



- Variance: Decreases gradually from PC1 (~2.2) to near-zero at PC7.
- Model Impact: PCA did not improve accuracy, so original features were kept.

Bayesian Statistical Methods





Naïve Bayes

Used as a baseline but assumes features are independent.

Bayesian GLM

Handles multicollinearity and provides stable, interpretable predictions.

Key Benefit: Gives probability-based risk scores for student dropout risk.

Bayesian Methods: Strengths & Limitations

- **Strength**: Uses prior knowledge and provides probability-based risk scores.
- **Limitation**: Can be computationally heavier and sensitive to prior assumptions.

Bayesian Decision-Making

- Accuracy after the threshold search: 60.13%
-
- Selected threshold: 0.50
- Baysian regression didn’t handle multicolinearity better than other models.

NAÏVE BAYES

```
> # Naive Bayes
> nb_model <- naiveBayes(as.factor(Completed_num) ~ Progress_Percentage + log_Time_Spent +
-                          Video_Completion_Rate + Quiz_Score_Avg +
-                          Days_Since_Last_Login + App_Usage_Percentage + Course_Level,
-                        data = train_data)
> nb_preds <- predict(nb_model, newdata = test_data)

> cat("Naive Bayes:", round(nb_acc, 2), "%\n")
Naive Bayes: 59.98 %
```

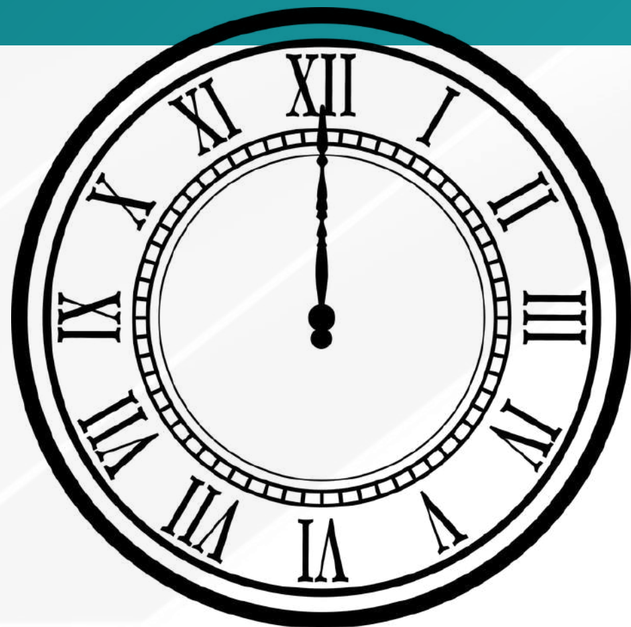
BAYESIAN REGRESSION

```
> # Bayesian Regression
> bayes_reg_model <- bayesglm(Completed_num ~ Progress_Percentage + log_Time_Spent +
+                               Video_Completion_Rate + Quiz_Score_Avg +
+                               Days_Since_Last_Login + App_Usage_Percentage + Course_Level,
+                             data = train_data, family = binomial(link = "logit"))
>
> bayes_preds <- predict(bayes_reg_model, newdata = test_data, type = "response")
> bayes_class <- ifelse(bayes_preds > 0.5, 1, 0)

> cat("Bayesian Regression:", round(bayes_acc, 2), "%\n")
Bayesian Regression: 60.13 %
```

Time Series Extension





- **Trend Analysis**

Track student engagement changes over time.

- **Seasonal Analysis**

Identify activity patterns during exams, holidays, and semesters.

- **Forecasting**

Use ARIMA to predict future completion and engagement trends.

Future System Enhancement

Business Benefit

Detecting declining engagement earlier and support timely interventions.



Industry Innovation Proposal





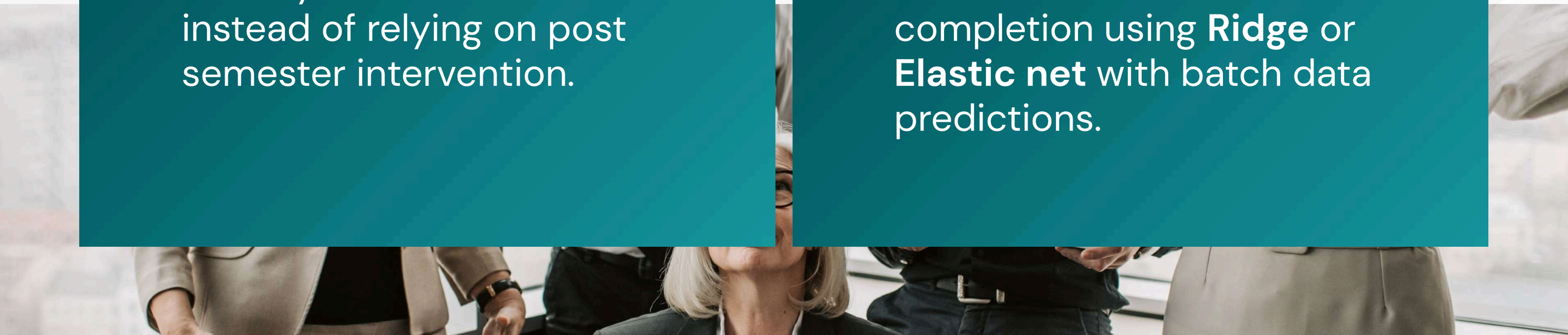
Proposed Solution

Business Need

- Identify at risk students earlier instead of relying on post semester intervention.

Proposed Solution

- Predict student course completion using **Ridge** or **Elastic net** with batch data predictions.



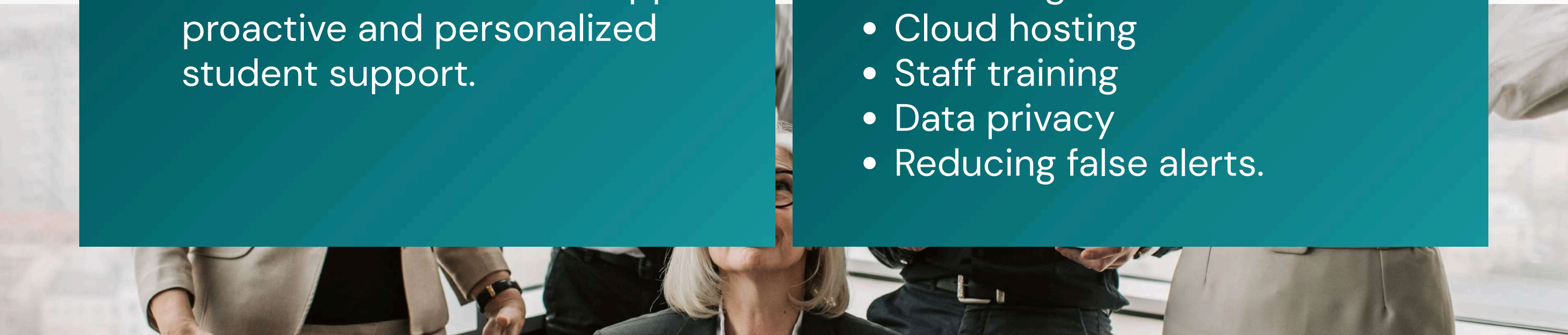
Proposed Solution

Benefits

- Provides risk scores to support proactive and personalized student support.

Requirements & Challenges

- LMS integration
- Cloud hosting
- Staff training
- Data privacy
- Reducing false alerts.



Industry Expert Validation





Expert Profile and Consultation Method

Expert: Minsandha Pathirana
Designation: Associate Data Engineer at Qoria
Consultation method: Through e-mail and whatsapp



Minsandha Pathirana
Associate Data Engineer
+94 703319289
Qoria

Main Feedback Received

- Model selection: Recommended interpretable regression models so academic advisors can understand student risk predictions.
- Data processing: Recommended a weekly automated batch-processing pipeline instead of real-time data streaming.
- Dashboard design: Emphasized a simple, intuitive advisordashboard with clear explanations of student risk levels.
- Expected impact: Improve early identification of at-risk students and support student retention.

Changes Made After Expert Feedback

Predictive modelling: Focused on interpretable regression-based models and compared multiple predictive approaches.

Early intervention: Used student progress, time spent, quiz scores, video completion, and inactivity indicators to identify potential risks.

A 70:30 training–testing split and accuracy evaluation to assess predictive performance.

Evidence

Minsandha Pathirana <minsandh4@gmail.com>
to me
Sep 21, 2026, 4:19 PM

Dear Project Team,

Thank you for walking me through your empirical modeling findings and the proposed EduPulse AI framework during our recent consultation session. I have reviewed your statistical architecture and deployment strategy, and I want to share my formal feedback to guide your next development phase.

The operational problem you are targeting is highly relevant; higher education institutions consistently struggle with reactive student retention management, often only identifying dropouts post-exams. Your approach to leveraging behavioral telemetry specifically progress percentages, time spent, and quiz scores directly addresses this core institutional pain point.

Here are my primary evaluations and recommendations for your deployment:

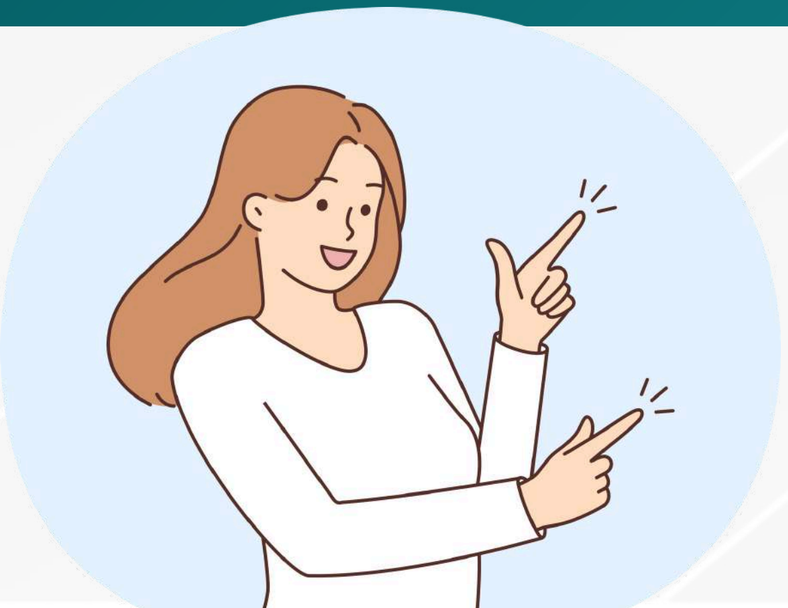
- **Model Selection and Interpretability:** I strongly endorse your decision to utilize Ridge Regression over complex, black-box machine learning algorithms. Academic advisors will simply not act on risk predictions unless they are provided with interpretable coefficients that explicitly explain why a specific student has been flagged.
- **Crucial Architectural Pivot (Batch vs. Real-Time):** While your implementation plan is highly feasible, I must offer a strong technical caveat regarding data ingestion. Attempting real-time streaming from legacy university LMS platforms (such as Moodle or Canvas) introduces unnecessary engineering overhead and risks database instability. I strongly advise modifying the EduPulse AI architecture to operate on a **weekly automated batch-processing pipeline**. This approach will minimize integration friction while ensuring risk scores remain fresh enough for actionable academic mentoring.
- **Dashboard UI and Impact:** Your automated early warning system, combined with an intuitive advisor dashboard, has the potential to significantly reduce administrative bottlenecks and improve persistence rates. Moving forward, keep the dashboard UI simple and ensure that the risk-tier explanations are immediately clear to the advisors.

You have built a very solid baseline predictive framework. Please ensure these architectural adjustments regarding the weekly batch scoring are integrated into your final implementation plan.

Let me know if you need further clarification on any of these points as you finalize the system.

Best regards,

Final Consultancy Recommendations





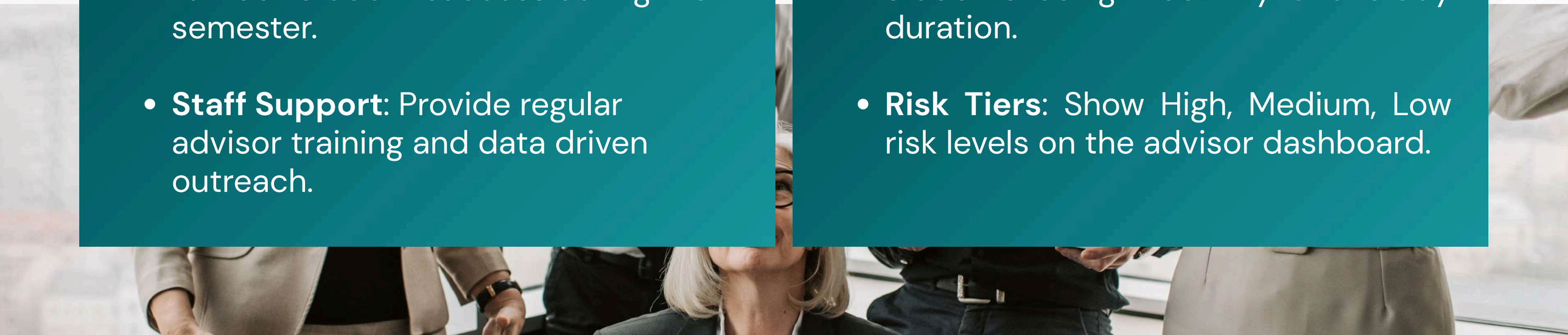
Recommendations

1. Strategic Recommendations

- **Proactive Support**: Use our analysis to track student success during the semester.
- **Staff Support**: Provide regular advisor training and data driven outreach.

2. Operational Recommendations

- **Weekly Predictions**: Identify at risk students using inactivity and study duration.
- **Risk Tiers**: Show High, Medium, Low risk levels on the advisor dashboard.



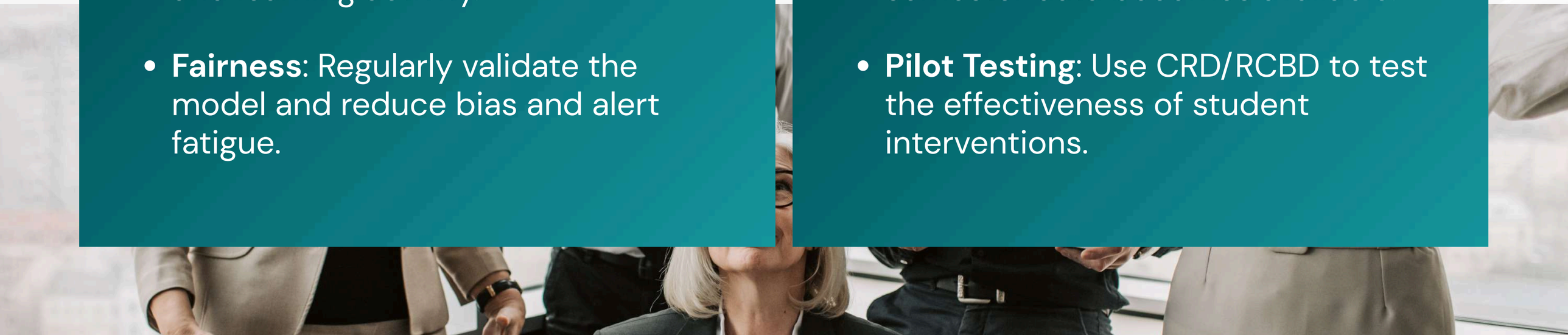
Recommendations

3. Risk & Ethics

- **Data Privacy**: Protect student data and learning activity.
- **Fairness**: Regularly validate the model and reduce bias and alert fatigue.

4. Future Research

- **Time Series:** Use ARIMA when multi semester data becomes available.
- **Pilot Testing**: Use CRD/RCBD to test the effectiveness of student interventions.



Project Conclusion

Summary of Project Performance



- Data: Converted 100,000 student records into an actionable retention framework.
- Key Finding: Study time and login gaps are important behavioral indicators of student success.
- Model & Deployment: Ridge + Bayesian GLM with weekly batch predictions provide interpretable risk scores.
- Future: Extend with time-series forecasting and controlled intervention experiments.



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THANK YOU

for Your Attention!



